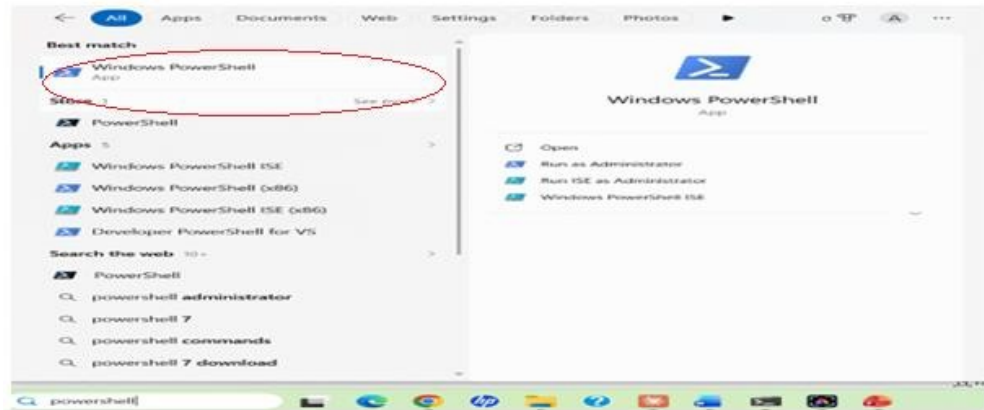


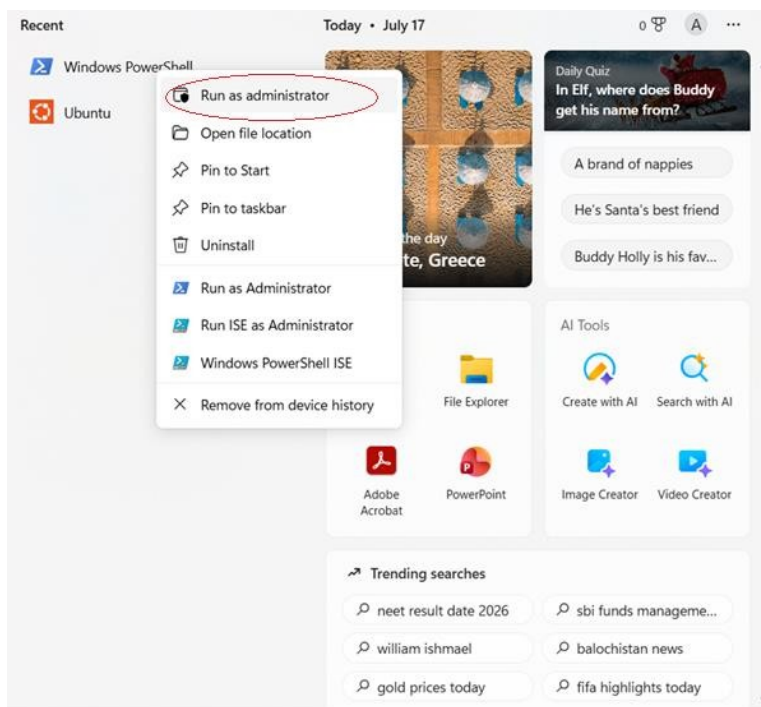
OSSP – SKILL 1

Installing Linux (WSL + Ubuntu), configuring the GCC compiler, and compiling a first C program.

Searched for Windows PowerShell in the Start menu.



Right-clicked Windows PowerShell and selected Run as administrator.



Ran `wsl --install` in PowerShell to install WSL along with Ubuntu, then restarted the PC when prompted.

```
Windows PowerShell
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PS C:\Users\samhi> wsl --install
```

After the restart, Ubuntu opened and asked me to create a UNIX username and password:

```
Enter new UNIX username: manu
New password: *****
Retype new password: *****
```

Updated the package lists with `sudo apt update` so everything is current.

```
manu@Manasvi: ~$ sudo apt update
[sudo: authenticate] Password:
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [424 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [549 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [97.9 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.7 kB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [159 kB]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.1 kB]
Get:11 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [361 kB]
Get:12 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [70.0 kB]
Get:13 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [128 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [98.0 kB]
Get:15 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [252 kB]
Get:16 http://archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [84.4 kB]
Get:17 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [187 kB]
Get:18 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Packages [3132 B]
Get:19 http://archive.ubuntu.com/ubuntu resolute-backports/universe Translation-en [7988 B]
Get:20 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [1056 B]
Fetched 2923 kB in 3s (959 kB/s)
Reading package lists... 7%
```

Installed the GCC C compiler using `sudo apt install build-essential` (it was already the newest version).

```
manu@Manasvi:~$ sudo apt install build-essential
build-essential is already the newest version (12.12ubuntu2.26.04.2).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 40
manu@Manasvi:~$
```

Checked the compiler with `gcc --version` — GCC 15.2.0 is installed and working.

```
manu@Manasvi:~$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

manu@Manasvi:~$ |
```

Created the project folder `ossp_shell` with `src`, `include`, and `obj` subfolders, then listed them with `ls`.

```
manu@Manasvi:~$ cd os_lab
manu@Manasvi:~/os_lab$ mkdir ossp_shell
manu@Manasvi:~/os_lab$ cd ossp_shell
manu@Manasvi:~/os_lab/ossp_shell$ mkdir src
manu@Manasvi:~/os_lab/ossp_shell$ mkdir include
manu@Manasvi:~/os_lab/ossp_shell$ mkdir obj
manu@Manasvi:~/os_lab/ossp_shell$ ls
include  obj  src
manu@Manasvi:~/os_lab/ossp_shell$ |
```

Wrote a simple C program in `src/hello.c` using `nano` that prints a message to the screen.

```
GNU nano 8.7.1 src/hello.c
#include <stdio.h>

int main() {
    printf("Hello,OSSP!GCC is configured\n");
    return 0;
}
```

Compiled the program directly with `gcc src/hello.c` and ran it with `./a.out`, which printed the output.

```
manu@Manasvi:~/os_lab/ossp_shell$ gcc src/hello.c
manu@Manasvi:~/os_lab/ossp_shell$ ./a.out
Hello,OSSP!GCC is configured
manu@Manasvi:~/os_lab/ossp_shell$ |
```

Created a Makefile with a simple rule that builds `hello` from `src/hello.c`.

```
GNU nano 8.7.1 Makefile *
hello:
    gcc src/hello.c -o hello|
```

Built the program with `make` (it reported `hello` was already up to date) and ran `./hello`, giving the same output.

```
manu@Manasvi:~/os_lab/oss_p_shell$ nano Makefile
manu@Manasvi:~/os_lab/oss_p_shell$ make
make: 'hello' is up to date.
manu@Manasvi:~/os_lab/oss_p_shell$ ./hello
Hello, OSSP!GCC is configured
manu@Manasvi:~/os_lab/oss_p_shell$ |
```

Understanding shell architecture

A shell runs in a simple loop: it reads a command, parses it, then creates a child process with `fork()` and runs the command with `exec()`, while the parent process waits (`wait()`) for the child to finish before showing the prompt again.